Problem 4

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$$F_n = \begin{cases} 1 & \text{if } n = 1 \\ 1 & \text{if } n = 2 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 3 \end{cases}$$

Proposition 1.

$$\sum_{j=1}^{m} F_{2j-1} = F_{2m}$$

Proof.
An impressive proof. □

Proposition 2.

$$\sum_{j=1}^{m} F_{2j} = F_{2m+1} - 1$$

Proof.
An impressive proof. □